

Year 1

AERO40005 - Materials 1

PROGRESS TEST n. 1 of 2

43%

Question 1

IMPORTANT INFORMATION (identical to what you will see in your real exam):

- Please submit a clear summary of all your answers to Question 1 as the first page of your exam script.
- For Parts (a) to (d) (multiple choice questions) you are required to submit your full written solution along with your chosen answer. Failure to do so, or submission of a wrong solution, will result in a mark of 0. The markers will check your procedure only for questions answered correctly.
- For part (e) (true or false questionnaire) you are not required to justify your answers. Note however that wrong answers will result in a negative mark, as detailed below.

Part (a)

A component is made from an isotropic metal with mechanical properties $E = 100$ GPa; $\nu = 0.3$; $\sigma_y = 300$ MPa. The stress state at a point in this component is given, in a certain Cartesian reference system, by

$$[\sigma] = \begin{pmatrix} 100 & 100 & 0 \\ 100 & 200 & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{ MPa.} = \begin{bmatrix} \sigma_{xx} & \tau_{xy} \\ \tau_{xy} & \sigma_{yy} \end{bmatrix}$$

The material obeys the Von Mises yield criterion.

- i. Is the given state of stress sufficient to initiate plasticity at this point?

Justify your answer. $\frac{100 + 200}{3} = 100 \text{ (MPa)} \neq \sigma_y$

A. Yes

☒ B. No $\sigma_{eq} = \sqrt{\frac{1}{2}[(\sigma_{xx} - \sigma_{yy})^2 + (\sigma_{xx} - \sigma_{zz})^2 + (\sigma_{yy} - \sigma_{zz})^2 + 6(\tau_{xy}^2 + \tau_{xz}^2 + \tau_{yz}^2)]} = 244.9 < \sigma_y$

C. The information given is not sufficient to answer the question.

0 [9%]

- ii. Calculate the volumetric strain in the component and state if this is

A. $\epsilon_v = 0$

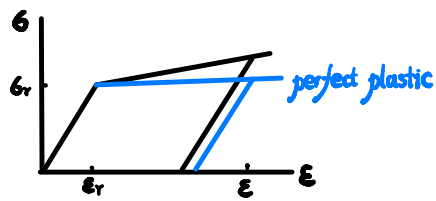
☒ B. $\epsilon_v = 1.2 \cdot 10^{-3}$

C. $\epsilon_v = 2.6 \cdot 10^{-3}$

D. None of the above.

$$\epsilon_v = \frac{\sigma_H}{K} = \frac{\frac{\sigma_H}{E}}{\frac{3(1-2\nu)}{E}} = 1.2 \times 10^{-3}$$

9 [9%]



Part (b)

A tensile specimen has gauge length $L_0 = 20$ mm. It is made from a perfectly plastic metal with nominal properties $E = 100$ GPa, $\sigma_y = 300$ MPa. During a test the gauge portion is stretched by $\Delta L = 2$ mm and the specimen is subsequently unloaded. Necking does not occur.

i. Calculate the final gauge length of the specimen and state if this is

A. $L_f = 21.32$ mm

☒ B. $L_f = 21.94$ mm

C. $L_f = 20.09$ mm

D. None of the above.

$$\epsilon_y = \frac{\sigma_y}{E} = 3 \times 10^{-3}$$

$$\epsilon = \frac{\Delta L}{L_0} = 0.1 \quad \therefore \Delta \epsilon = 0.047$$

$$\therefore \Delta L = \Delta \epsilon H \quad L = L_0 + \Delta L = L_0 + L_0 \Delta \epsilon = 21.94 \text{ mm}$$

0 [18%]

ii. Calculate the hardness of the material after this test and state if this is

A. 100 MPa

B. 900 MPa

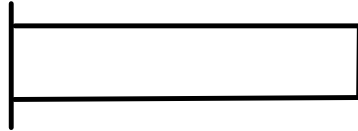
C. > 900 MPa

D. None of the above.

$$H = 3\sigma_y = 900 \text{ MPa}$$

(Tabor's construction)

0 [8%]

Part (c)

A horizontal cantilever beam of length $L_0 = 3 \text{ m}$ and square cross-section of area $t \times t$ needs to be designed. The beam, of density ρ , is required to carry only its own self-weight without yielding. The maximum stress for a cantilever subject to its own self-weight is given by

$$\sigma_{\max} = \frac{3\rho g L^2}{t},$$

where $g = 9.81 \text{ m/s}^2$ is the acceleration due to gravity.

Calculate the merit index for a design of minimum cost and state if this is

A $\sigma_Y^2 / (\rho^3 c)$

B. $\sigma_Y / (\rho c)$

C. $\sigma_Y^2 / (\rho c)$

D. None of the above.

cost = mc

$$\sigma_{\max} = \frac{3\rho g L^2}{t} \quad \therefore t = \frac{3\rho g L^2}{\sigma_Y}$$

$$m = \rho V = \rho t L_0 = \rho \frac{3\rho g L^2}{\sigma_Y} L_0$$

$$\therefore m \propto \frac{\rho^2}{\sigma_Y}$$

$$\therefore mc \propto \frac{\rho^2 c}{\sigma_Y}$$

$$\therefore M_i = \frac{\sigma_Y^2}{\rho^3 c}$$

18 [18%]

Part (d)

A resistance strain gauge of gauge factor $S = 2$ is inserted in a Wheatstone bridge in half-bridge configuration. If the strain gauge is subject to a strain of $\varepsilon = 10^{-3}$ and its original resistance is $R_0 = 120 \Omega$, calculate the expected change in resistance ΔR and state if this is

A. $\Delta R = 2.304 \Omega$

B. $\Delta R = 1.152 \Omega$

C. $\Delta R = 0.002 \Omega$

☒ D None of the above

$$\frac{\Delta R}{R} = S\varepsilon$$

$$\therefore \Delta R = S\varepsilon R = 0.24 \Omega$$

0 [6%]

Part (e)

State if the following are true or false. You will receive 8 points for each correct answer, -8 points for each incorrect answer, 0 points for each unanswered question.

~~F~~ i. The formula $\sigma_t = \sigma_n (1 + \varepsilon_n)$ is only valid for metallic materials.

~~F~~ ~~T~~ ii. The indentation of a solid by Rockwell, Brinell or Vickers indenters gives rise to self-similar strain distributions.

~~T~~ iii. A body is in equilibrium, subject to vanishing body forces ($\underline{b} = 0$) and in a state of plane stress. If σ_{xx} does not depend on the x coordinate, then τ_{xy} does not depend on the y coordinate.

~~T~~ iv. A body is subject to a state of equibiaxial plane stress, that is $\sigma_{xx} = \sigma_{yy} \neq 0$ while all other stress components are equal to zero. The state of stress in the $(x-y)$ plane is, in this case, independent of the choice of reference system.

16 [32%]